

Samit Madatanapalli

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EXPERIENCE

Honeywell <i>Intern Software Engineer</i>	05/2025 – 08/2025 <i>Pittsburgh, PA</i>
- Developed and deployed a real-time free-fall detection feature in C for Honeywell embedded Linux devices; conducted industrial drop-testing to validate sensor processing accuracy - Reverse-engineered legacy battery charger systems by analyzing MCU-to-battery communication (I2C, UART, 1-Wire) using Saleae logic analyzers, Perl scripts, and DMM, ensuring hardware compatibility for next-generation charging systems - Engineered Python/Shell scripts to visualize Adaptive Frequency Hopping (AFH) data; generated 2D channel maps to resolve critical Bluetooth interference issues for major clients. - Built a Python-based telemetry tool to automate the extraction and visualization of performance metrics, reducing manual debugging time for the electrical engineering team.	
Flourish: Grow with Self-Care (Startup) <i>Intern Full Stack Mobile Application Engineer</i>	01/2025 – 05/2025 <i>Remote</i>
- Built a "Plants Encyclopedia" feature using React Native, implementing a real-time search bar and custom UI animations for the user interface - Integrated Appwrite and Python services to store plant metadata and user preferences, ensuring the app displays personalized content correctly	
Persistent Systems <i>Intern Data Engineer Mentee</i>	05/2024 – 08/2024 <i>Remote</i>
- Refactored a personal AI Fitness Tracker using ER/Studio and SQL, implementing normalized database schemas to optimize storage for health metrics and AI-generated nutritional data. - Applied enterprise-grade data governance standards to ensure data integrity and query efficiency; collaborated with mentors in daily syncs to refine data transformation workflows.	

PROJECTS

Esotaira: Military Omnidirectional UAV (Startup) <i>Capstone Embedded Software Lead</i>	08/2025 – Present <i>University Park, PA</i>
- Developed C++ firmware to manage 0–180° tilt-rotor modulation, enabling omnidirectional thrust and high-agility maneuvering for military-spec flight profiles - Engineered the full electrical architecture, implementing high-reliability wire routing and signal-integrity hardening (soldering, heat-shrinking) for ruggedized use - Optimized Onyx 3D-printed airframe components and assisted in the construction of an epoxy-glass testing enclosure to ensure structural integrity during high-torque stress tests	
Offline AI-enabled Drone <i>Embedded Lead</i>	09/2025 – Present <i>University Park, PA</i>
- Engineered a custom quadrotor drone that broadcasts live aerial telemetry to a localized Raspberry Pi 5 ground station for high-speed field surveillance. - Developed an offline detection pipeline using YOLOv8 and OpenCV to identify personnel and vehicles in real-time, overlaying dynamic AI labels onto a mobile touchscreen.	

EDUCATION

The Pennsylvania State University College of Engineering	Graduated: 05/2026 <i>University Park, PA</i>
- B.S. in Computer Engineering - Dean's List 2023-2024-2025	
	GPA: 3.50